

AP[®] STATISTICS FORMULA & INFERENCE DECISION GUIDE

Stop Memorizing. Master It

You do not need another dense 600-page prep book. Designed for high-yield clarity, this companion cuts through textbook

Inside This Hardcover Edition:

- * Plain-English Formulas: Every parameter defined
- * Unit Mindmaps: Visual 1-page concept roadmaps
- * Master Mnemonics: Zero-guesswork checklist
- * FRQ Sentence Frames: Standardized phrasing
- * Top 25 Traps: Never confuse 'failing to reject' with 'accepting'
- * Exam Pacing Strategy: Pacing blueprints for every section
- * Free Digital Companion: Instant interactive formulas and graphs

AP STATISTICS FORMULA & INFERENCE DECISION GUIDE BOOK 1

[KDP BARCODE ZONE]

FORMULAS

$\bar{x}$ Mean σ Standard Deviation n Sample Size

$$\bar{x} = \frac{\sum x + \mu^2}{n}$$

$$\sigma = \frac{\sigma}{n\sqrt{n}}$$

p Probability

Probability

$$p = \frac{x-1}{n}$$

Probability

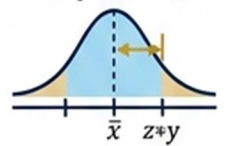
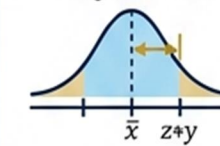
CONFIDENCE INTERVALS

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$



Margin of Error

Margin of Error



SIGNIFICANCE TESTS

p-values

Null Hypothesis

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

Test Statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Alternative Hypotheses

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

$$-\sqrt{n}$$

Test Statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

INFERENCE DECISION TREE



Compare Groups?

Yes

No

Categorical Groups?

Categorical or Quantitative Data?

Yes

No

Yes

No

z-test Test

t-test Test

Chi-square Test

Regression